Diploma in Engineering and Technology • Semester 1

Applied Physics-I Downloadable Notes

Student-friendly digital textbook + workbook + revision guide + answer guide. Only the selected section loads on screen. The PDF button opens browser Save as PDF without freezing the page.

Course Overview

Measurements → Vectors → Motion → Rotation → Energy & Heat → Elasticity → Fluids.

Course objectives

- Understand important physical principles used in engineering.
- Develop numerical problem-solving skill.
- Apply mechanics, energy, heat and fluid concepts to practical situations.

Course outcomes

- CO1: Apply laws of mechanics in recoil of gun and rocket propulsion.
- CO2: Apply circular and rotational motion concepts.
- CO3: Solve problems involving work, energy, power, temperature and friction.
- CO4: Use fluid dynamics in atomiser and airfoil.

Study method

Daily method

1. Read one lesson.
2. Copy formulas with units.
3. Redraw the diagram.
4. Solve worked examples.

Numerical format

1. Given Data
2. Formula Used
3. Substitution
4. Calculation
5. Final Answer with unit

Long answer format

1. Definition

2. Explanation
3. Labelled diagram
4. Application + example

Module map

Mini Formula Bank

Search formulas, topics and units. This section loads only when selected.

Search formula, topic or unit...

Module 1

Module 1: Measurements, Vectors and Laws of Motion

CO1: Apply laws of mechanics in rocket propulsion and recoil of gun. Only this selected module is loaded.

Search inside selected module...

Module 1 • 9 hours

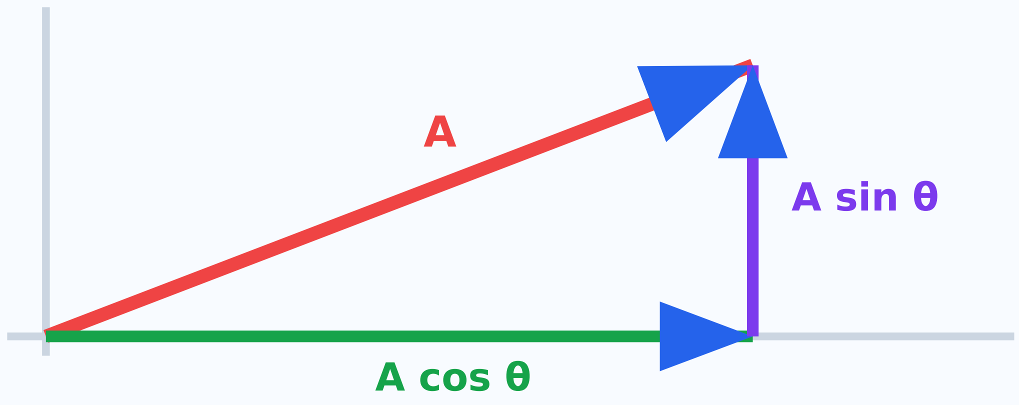
Measurements, Vectors and Laws of Motion

CO1: Apply laws of mechanics in rocket propulsion and recoil of gun.

□ module-□ units, errors, vectors, Newton's laws, momentum, impulse, recoil of gun, rocket propulsion □□□□□□ exam point of view-□ □□□□□□□□.

Visual diagram

Vector Resolution



Learning outcomes

- Physical quantities, units and systems
- Errors in measurement
- Scalar and vector quantities
- Equations of motion and Newton's laws
- Momentum, impulse, recoil and rocket propulsion

Physical quantities, units and systems

Lesson 1

ENGLISH

A physical quantity is anything that can be measured and expressed using a numerical value and a unit. Fundamental quantities are independent quantities such as length, mass and time. Derived quantities are obtained by combining fundamental quantities, for example velocity, acceleration, force and work.

MALAYALAM HELPER

□□□□□□□ □□□□□□□□ □□□ quantity □□□ physical quantity. □□□□□ numerical value □□ unit □□ □□□□. Length, mass, time □□□□□□□□ fundamental quantities □□□. Velocity, acceleration, force □□□□□□□□ derived quantities □□□.

Physical quantity = numerical value × unit | **$1\ N = 1\ kg\ m\ s^{-2}$** | **$1\ J = 1\ N\ m$**

Key points

- Use SI units in engineering answers.
- Write unit conversion clearly before substitution.
- Never write a numerical final answer without unit.

Application

Engineering drawings, laboratory readings and machine design calculations need consistent units.

> **Study tip:** Number without unit is incomplete.

Errors in measurement

Lesson 2

ENGLISH

Error is the difference between measured value and true or accepted value. Systematic error is repeated in the same direction due to a faulty instrument or method. Random error changes unpredictably during repeated measurements.

MALAYALAM HELPER

Measured value-□□□ true/accepted value-□□□ □□□□□□□□□ □□□□□□□□□□□□ error. Faulty instrument □□□□□□□□□□ wrong method □□□□□ systematic error □□□□□□□□. Repeated reading-□ unpredictable variation □□□□□□ random error □□□□□□□□.

Absolute error = |measured − true|

Relative error = absolute error / true value

% error = relative error × 100

Key points

- Take repeated readings and mean value.
- Avoid parallax error while reading scale.
- Percentage error helps compare different measurements.

Application

Quality inspection, machining tolerance and laboratory experiments.

Study tip: Write absolute error with unit; relative error has no unit.

Scalar and vector quantities

Lesson 3

ENGLISH

A scalar quantity has magnitude only. A vector quantity has magnitude and direction. Temperature, mass and time are scalars. Displacement, velocity, acceleration, force and momentum are vectors. Vector resolution changes one vector into rectangular components.

MALAYALAM HELPER

Scalar quantity-□□□□ magnitude □□□□□□. Vector quantity-□□□□ magnitude □□□□ direction □□ □□□□. Temperature, mass, time scalar □□□. Displacement, velocity, acceleration, force, momentum vector □□□.

$A_x = A \cos \theta$ | **$A_y = A \sin \theta$** | **$R = \sqrt{(A^2 + B^2)}$**

Key points

- Represent vector by an arrow.
- Arrow length shows magnitude.
- Arrow head shows direction.

Application

Crane lifting, bridge forces, slope motion and surveying.

Study tip: Perpendicular vectors form a right triangle.

Equations of motion and Newton's laws

Lesson 4

ENGLISH

Equations of motion connect displacement, velocity, acceleration and time for uniform acceleration. Newton's laws explain inertia, force and action-reaction. Force is the rate of change of momentum; for constant mass it becomes F = ma.

MALAYALAM HELPER

Uniform acceleration □□□□ motion-□ displacement, velocity, acceleration, time □□□□□□□□□□ □□□□□□□□□ equations of motion. Newton's laws inertia, force, action-reaction explain □□□□□□□□□□. Constant mass □□□□□□□□ F = ma.

$v = u + at$ **$s = ut + \frac{1}{2}at^2$** **$F = ma$**

Key points

- First law explains inertia.
- Second law gives force equation.
- Third law states action and reaction are equal and opposite.

Application

Vehicle acceleration, braking distance and machine motion analysis.

Study tip: Use equations of motion only when acceleration is constant.

Momentum, impulse, recoil and rocket propulsion

Lesson 5

ENGLISH

Momentum is the product of mass and velocity. Impulse is the product of force and time and equals change in momentum. Recoil of gun and rocket propulsion are based on conservation of linear momentum. In a rocket, high-speed gases move backward and the rocket moves forward.

MALAYALAM HELPER

Momentum = mass × velocity. Impulse = force × time = change in momentum. Recoil of gun, rocket propulsion □□□□□□ conservation of linear momentum

□□□□□□□□□□□□□□□□□. Rocket-□ gases backward □□□□□□□□□ rocket forward move □□□□□□□.

$p = mv$ **$J = Ft = \Delta p$** **$m_1v_1 + m_2v_2 = \text{constant}$**

Key points

- Total momentum before and after interaction remains constant if external force is zero.
- Gun moves backward after bullet moves forward.
- Rocket does not need air to push against.

Application

Rockets, jet aircraft, pile driving and safety airbags.

Study tip: Impulse increases when stopping time increases; this reduces force.

Solved Problems

Exam-friendly method: Given Data → Formula Used → Substitution → Final Answer.

Problem 1

A force of 4 N acts on a 3 kg body. Find acceleration.

Given $F = 4$ N, $m = 3$ kg. Formula $a = F/m$. Therefore $a = 4/3 = 1.33$ m s^{-2}.

Problem 2

A 2 kg body moves at 5 m/s. Find momentum.

$p = mv = 2 \times 5 = 10$ kg m s^{-1}.

Problem 3

A force of 10 N acts for 0.5 s. Find impulse.

$J = Ft = 10 \times 0.5 = 5$ N s.

Expected Questions

Questions only are shown inside modules. Answers / hints are kept only in the Master Answer Key tab.

Q1. Define physical quantity.

Q2. What is vector resolution?

Q3. State Newton's second law.

Q4. Why does a gun recoil?

Q5. Explain rocket propulsion.

Long Answer Writing Practice

Use this structure for five-mark and ten-mark questions. Full answer checking is kept in the Master Answer Key tab.

Explain conservation of linear momentum with recoil of gun.

- State conservation law.
- Before firing, total momentum is zero.
- After firing, bullet momentum forward equals gun momentum backward.
- This backward motion of gun is recoil.
- Add formula and application.

Writing hint: Use the above points to build the answer in your own words.

1 mark
Definition or formula.

2 marks
Definition + one point.

5 marks
Statement + diagram + application.

10 marks
Detailed explanation + derivation + diagram.

Module 2

Module 2: Circular and Rotational Motion

CO2: Apply concepts of circular and rotational motion. Only this selected module is loaded.

Search inside selected module...

Module 2 • 9 hours

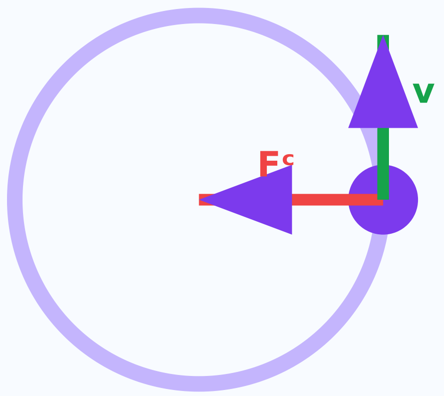
Circular and Rotational Motion

CO2: Apply concepts of circular and rotational motion.

Circular motion, angular quantities, centripetal force, banking of roads, moment of inertia, torque, angular momentum □□□□□□ □ module-□ □□□□□□□□.

Visual diagram

Circular Motion



Centripetal force acts towards the centre

Learning outcomes

- Angular displacement, velocity and acceleration
- Relation between linear and angular quantities
- Centripetal force and acceleration
- Banking of roads
- Moment of inertia, torque and angular momentum

Angular displacement, velocity and acceleration

Lesson 1

ENGLISH

Angular displacement is the angle swept by a rotating body. Angular velocity is rate of change of angular displacement. Angular acceleration is rate of change of angular velocity. Radian is used as the standard unit for angle in physics calculations.

MALAYALAM HELPER

Rotating body sweep □□□□□□□□□□ angle □□□ angular displacement. Angular displacement-□□□□ rate □□□ angular velocity. Angular velocity-□□□□ rate □□□ angular acceleration. Calculation-□ radian □□□□□□□□□□□□□□□□.

$\omega = \theta/t$ | $\alpha = (\omega - \omega_0)/t$ | $1\ \text{rev} = 2\pi\ \text{rad}$

Key points

- One revolution = 2π rad.
- Angular velocity is measured in rad s^{-1}.
- Angular acceleration is measured in rad s^{-2}.

Application

Fans, wheels, turbines and rotating shafts.

Study tip: Convert rpm to rad/s before numerical problems.

Relation between linear and angular quantities

Lesson 2

ENGLISH

A point on a rotating body has both angular motion and linear motion. Linear velocity depends on radius and angular velocity. Points farther from the centre have greater linear velocity for the same angular velocity.

MALAYALAM HELPER

Rotating body-□ □□□□ point-□□□□ angular motion □□ linear motion □□ □□□□□□□□□. Linear velocity radius-□□□ angular velocity-□□□□□ proportional □□□. Centre-□ □□□□□□ □□□□ □□□□□□□□□ linear velocity □□□□□.

$s = r\theta$ | $v = r\omega$ | $a = r\alpha$

Key points

- All points in a rigid body have same angular velocity.
- Outer points travel more distance in same time.
- Tangential acceleration is related to angular acceleration.

Application

Wheel rims, pulley belts and gear systems.

Study tip: The letter r connects circular and linear quantities.

Centripetal force and acceleration

Lesson 3

ENGLISH

A body moving in a circle needs acceleration towards the centre even if speed is constant. This acceleration is centripetal acceleration. The force producing it is centripetal force. Without this force, the body moves tangentially.

MALAYALAM HELPER

Constant speed-□ circle-□ move □□□□□□□□□ direction □□□□□□□□□□□□ acceleration centre direction-□ □□□□□□□□. □□□□□ centripetal acceleration. □□□□□ □□□□□□□□ force centripetal force □□□.

$a^c = v^2/r$ $a^c = r\omega^2$ $F^c = mv^2/r$

Key points

- Direction is always towards centre.
- It changes direction of velocity.
- It is not a new force; it is provided by tension, friction, gravity etc.

Application

Vehicle turning, satellites, stone tied to a string.

Study tip: Centripetal means centre-seeking.

Banking of roads

Lesson 4

ENGLISH

Banking means raising the outer edge of a curved road. It provides the required centripetal force safely while vehicles take a turn. Proper banking reduces dependence on friction and improves safety at curves.

MALAYALAM HELPER

Curved road-□□□□ outer edge □□□□□□□□□□□□□□ banking. Vehicle turn □□□□□□□□□□□ □□□□□ centripetal force safely provide □□□□□□□ □□□ □□□□□□□□□□□□□. Banking friction dependence □□□□□□□□□□□□□.

$\tan\theta = v^2/(rg)$ **$v = \sqrt{rg\tan\theta}$**

Key points

- Outer edge is higher than inner edge.
- Useful on highways and railway tracks.
- Design depends on speed and radius of curve.

Application

Road curves, railway tracks and race tracks.

Study tip: Higher design speed needs greater banking angle.

Moment of inertia, torque and angular momentum

Lesson 5

ENGLISH

Moment of inertia is the rotational analogue of mass. It depends on mass and distribution of mass about the axis of rotation. Torque is turning effect of force. Angular momentum is the product of moment of inertia and angular velocity.

MALAYALAM HELPER

Moment of inertia rotational motion-□□ mass □□□□□□□□. Mass □□□□□□ axis-□ □□□□□□ distribute □□□□□□□□□□□□□□□□ □□□□□□□□ □□□□□□□□□□□. Torque force-□□□□ turning effect □□□. Angular momentum = Iω.

$I = \Sigma mr^2$ | $\tau = I\alpha$ | $L = I\omega$

Key points

- More mass away from axis gives more moment of inertia.
- Torque produces angular acceleration.
- Angular momentum is conserved when external torque is zero.

Application

Flywheels, turbines, gyroscopes and rotating machines.

Study tip: Moment of inertia increases strongly with radius because of r^2.

Solved Problems

Exam-friendly method: Given Data → Formula Used → Substitution → Final Answer.

Problem 1

A wheel has radius 0.5 m and angular velocity 4 rad/s. Find linear velocity.

$v = r\omega = 0.5 \times 4 = 2$ m/s.

Problem 2

A 2 kg body moves in a circle of radius 1 m with speed 4 m/s. Find centripetal force.

$F = mv^2/r = 2 \times 16 / 1 = 32$ N.

Problem 3

A body has $I = 3$ kg m² and $\omega = 5$ rad/s. Find angular momentum.

$L = I\omega = 3 \times 5 = 15$ kg m^2 s^{-1}.

Expected Questions

Questions only are shown inside modules. Answers / hints are kept only in the Master Answer Key tab.

Q1. Define angular velocity.

Q2. What is centripetal force?

Q3. Why are roads banked?

Q4. Define moment of inertia.

Q5. State parallel axis theorem.

Long Answer Writing Practice

Use this structure for five-mark and ten-mark questions. Full answer checking is kept in the Master Answer Key tab.

Explain banking of roads.

- Define banking.
- Draw a vehicle on a banked curve.
- Mention centripetal force requirement.
- Write formula $\tan\theta = v^2/(rg)$.
- Give road and railway applications.

Writing hint: Use the above points to build the answer in your own words.

1 mark
Definition or formula.

2 marks
Definition + one point.

5 marks
Statement + diagram + application.

10 marks
Detailed explanation + derivation + diagram.

Module 3

Module 3: Work, Energy, Power, Friction and Heat

CO3: Solve problems involving work, energy, power, temperature and friction. Only this selected module is loaded.

Search inside selected module...

Module 3 • 9 hours

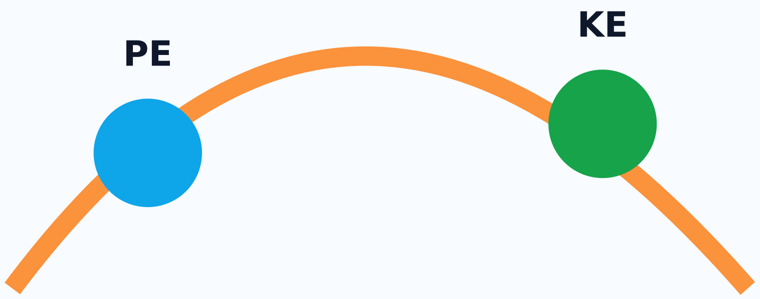
Work, Energy, Power, Friction and Heat

CO3: Solve problems involving work, energy, power, temperature and friction.

Work, friction, energy, power, conservation of energy, heat, temperature, thermometry, heat transfer □□□□□□ □ module-□ □□□□□□□□.

Visual diagram

Energy Transformation



Potential energy changes into kinetic energy

Learning outcomes

- Work and friction
- Energy and conservation of energy
- Power and efficiency
- Heat, temperature and thermometry

- Modes of heat transfer

Work and friction

Lesson 1

ENGLISH

Work is done when a force produces displacement. Work depends on force, displacement and angle between them. Friction is a resistive force that opposes relative motion. Static friction acts before motion starts and kinetic friction acts during motion.

MALAYALAM HELPER

Force displacement produce □□□□□□□ work done □□□□□ □□□□□□□□□. Work force, displacement, angle □□□□□□□□ □□□□□□□□□□□□. Relative motion-□□ oppose □□□□□□□□□□ resistive force □□□ friction. Motion start □□□□□□□□□□□□□ □□□□□□□ static friction, motion □□□□□□□□□□□ kinetic friction.

$W = Fs \cos \theta$ $F = \mu N$ $N = mg$

Key points

- No displacement means no mechanical work.
- Friction helps walking and braking.
- Friction wastes energy as heat in machines.

Application

Brakes, clutches, belt drives and lubrication.

Study tip: Friction is useful and harmful depending on situation.

Energy and conservation of energy

Lesson 2

ENGLISH

Energy is the capacity to do work. Kinetic energy is energy due to motion. Potential energy is energy due to position. According to conservation of energy, energy can neither be created nor destroyed; it can only be transformed from one form to another.

MALAYALAM HELPER

Work □□□□□□□□□□□ capacity □□□ energy. Motion □□□□□ □□□□ energy kinetic energy. Position/height □□□□□ □□□□ energy potential energy. Conservation of energy □□□□□□□ energy create □□□□□□□□ destroy □□□□□□□□ □□□□□□□□; □□□ form-□ □□□□□□ □□□□□□□ form-□□□□□□ □□□□□.

$KE = \frac{1}{2}mv^2$ **$PE = mgh$** **Total energy = constant**

Key points

- Falling body converts PE into KE.
- Machines convert energy into useful work.
- Losses usually appear as heat and sound.

Application

Hydroelectric power, cranes, lifts and machine tools.

Study tip: In falling body problems, PE decreases and KE increases.

Power and efficiency

Lesson 3

ENGLISH

Power is the rate of doing work or rate of energy conversion. A machine with higher power performs the same work in less time. Efficiency compares useful output energy or power with input energy or power.

MALAYALAM HELPER

Work □□□□□□□□□□□□□□□ rate □□□□□□□□□□ energy conversion-□□□□ rate □□□ power. Higher power machine same work □□□□□□□ time-□ □□□□□□□. Efficiency useful output-□□ input-□□□□□ compare □□□□□□□□□□.

$P = W/t$ **$P = E/t$** **Efficiency = output/input × 100**

Key points

- Power unit is watt.
- Commercial unit of electrical energy is kWh.
- Efficiency is always less than 100% for practical machines.

Application

Motors, pumps, lifts and power plants.

Study tip: 1 horsepower = 746 W.

Heat, temperature and thermometry

Lesson 4

ENGLISH

Heat is energy transferred because of temperature difference. Temperature indicates degree of hotness or coldness. Thermometers measure temperature using a physical property that changes uniformly with temperature. Mercury thermometer and pyrometer are common examples.

MALAYALAM HELPER

Temperature difference □□□□□ transfer □□□□□□□□□ energy □□□ heat. Temperature hot/cold level □□□□□□□□□□□□. Temperature change □□□□□□□□□□ physical property uniformly □□□□□□□□□ □□□□□□□□□□ thermometer measure □□□□□□□□□□.

$Q = mc\Delta T$ **$K = {}^\circ C + 273$** **${}^\circ F = 1.8{}^\circ C + 32$**

Key points

- Heat unit is joule.
- Temperature unit is kelvin in SI.
- Pyrometer measures high temperature without contact.

Application

Furnaces, boilers, engine cooling and heat exchangers.

Study tip: Do not write °K; correct SI symbol is K.

Modes of heat transfer

Lesson 5

ENGLISH

Heat is transferred by conduction, convection and radiation. Conduction needs material contact. Convection occurs through bulk movement of fluid. Radiation transfers heat by electromagnetic waves and can travel through vacuum.

MALAYALAM HELPER

Heat transfer conduction, convection, radiation □□□□□ modes □□□ □□□□□□□□□□. Conduction-□□ contact □□□□. Convection fluid movement □□□. Radiation vacuum-□ □□□□ travel □□□□□□□□.

$Q = mc\Delta T$ | **Heat flows from high temperature to low temperature**

Key points

- Metal spoon gets hot by conduction.
- Sea breeze is due to convection.
- Sunlight reaches Earth by radiation.

Application

Insulation, furnaces, radiators and solar heating.

Study tip: Radiation does not need a medium.

Solved Problems

Exam-friendly method: Given Data → Formula Used → Substitution → Final Answer.

Problem 1

A force of 20 N moves a body 6 m in same direction. Find work.

$W = Fs = 20 \times 6 = 120$ J.

Problem 2

Find KE of 2 kg body moving at 5 m/s.

$KE = \frac{1}{2}mv^2 = 0.5 \times 2 \times 25 = 25$ J.

Problem 3

Calculate power if 400 J work is done in 4 s.

$P = W/t = 400/4 = 100$ W.

Problem 4

Convert 100°C to Fahrenheit.

$F = 1.8C + 32 = 180 + 32 = 212$°F.

Expected Questions

Questions only are shown inside modules. Answers / hints are kept only in the Master Answer Key tab.

Q1. Define work.

Q2. What is friction?

Q3. State conservation of energy.

Q4. Define power.

Q5. Name modes of heat transfer.

Long Answer Writing Practice

Use this structure for five-mark and ten-mark questions. Full answer checking is kept in the Master Answer Key tab.

Explain modes of heat transfer with examples.

- Define heat transfer.

- Explain conduction with example.
- Explain convection with example.
- Explain radiation with example.
- Add engineering applications.

Writing hint: Use the above points to build the answer in your own words.

1 mark
Definition or formula.

2 marks
Definition + one point.

5 marks
Statement + diagram + application.

10 marks
Detailed explanation + derivation + diagram.

Module 4

Module 4: Elasticity, Fluids and Bernoulli Applications

CO4: Use fluid dynamics in atomiser and airfoil. Only this selected module is loaded.

Search inside selected module...

Module 4 • 9 hours

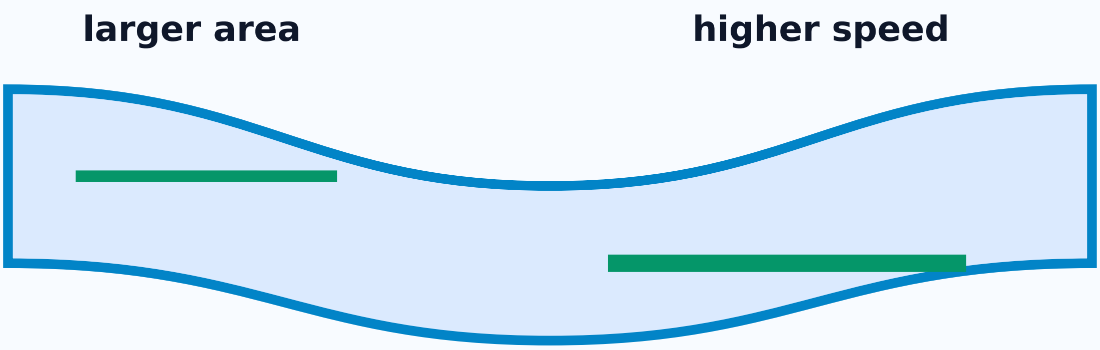
Elasticity, Fluids and Bernoulli Applications

CO4: Use fluid dynamics in atomiser and airfoil.

Elasticity, stress, strain, pressure, surface tension, viscosity, continuity equation, Bernoulli theorem, airfoil, atomiser □□□□□□ □□□□□□□□□.

Visual diagram

Continuity and Bernoulli



Narrow region: velocity increases, pressure decreases

Learning outcomes

- Elasticity, stress and strain
- Pressure and fluids
- Surface tension and capillarity
- Viscosity and Reynolds number
- Continuity equation and Bernoulli theorem

Elasticity, stress and strain

Lesson 1

ENGLISH

Elasticity is the property by which a material regains its original shape and size after removal of deforming force. Stress is restoring force per unit area. Strain is the ratio of change in dimension to original dimension. Young's modulus is stress divided by strain.

MALAYALAM HELPER

Deforming force remove □□□□□□□ original shape and size regain □□□□□□□□□□□ property □□□ elasticity. Stress = restoring force per unit area. Strain = change in dimension / original dimension. Young's modulus = stress / strain.

Stress = F/A **Strain = ΔL/L** **Y = stress/strain**

Key points

- Stress has unit pascal.
- Strain has no unit.
- Hooke's law is valid within elastic limit.

Application

Springs, bridges, beams, wires and machine components.

Study tip: Stress has unit; strain has no unit.

Pressure and fluids

Lesson 2

ENGLISH

Pressure is force per unit area. In fluids, pressure acts normally on any surface. Gauge pressure is pressure above atmospheric pressure. Absolute pressure is the sum of atmospheric pressure and gauge pressure.

MALAYALAM HELPER

Pressure = force per unit area. Fluid pressure surface-□ normal direction-□ act □□□□□□□. Gauge pressure atmospheric pressure-□□□□□□□ □□□□□ □□□□ pressure □□□. Absolute pressure = atmospheric pressure + gauge pressure.

$P = F/A$ | $P = \rho gh$ | $P_{abs} = P_{atm} + P_{gauge}$

Key points

- Pressure unit is pascal.
- Smaller area gives greater pressure for same force.
- Fluid pressure increases with depth.

Application

Hydraulic machines, dams, pressure gauges and pumps.

Study tip: Use absolute pressure when total pressure is required.

Surface tension and capillarity

Lesson 3

ENGLISH

Surface tension is the property of a liquid surface due to which it behaves like a stretched elastic membrane. Capillarity is the rise or fall of liquid in a narrow tube due to surface tension and adhesive/cohesive forces.

MALAYALAM HELPER

Liquid surface stretched membrane □□□□ behave □□□□□□□□□□□ property □□□ surface tension. Narrow tube-□ liquid rise/fall □□□□□□□□□□□ capillarity □□□. Surface tension, adhesive/cohesive forces □□□□□ □□□ □□□□□□□□□□□□□□.

$h = 2T \cos\theta / (r\rho g)$ | **Surface tension unit = N/m**

Key points

- Small drops are nearly spherical due to surface tension.
- Capillary rise increases when tube radius decreases.
- Meniscus shape depends on liquid and tube material.

Application

Ink pens, wicks, soil water movement and medical capillary tubes.

Study tip: Narrower capillary tube gives greater rise.

Viscosity and Reynolds number

Lesson 4

ENGLISH

Viscosity is the internal resistance offered by a fluid to relative motion between its layers. A highly viscous liquid flows slowly. Reynolds number is a dimensionless number used to identify laminar or turbulent flow.

MALAYALAM HELPER

Fluid layers □□□□□□□□□□ relative motion-□□ oppose □□□□□□□□□ internal resistance □□□ viscosity. Highly viscous liquid slow □□□ flow □□□□□□□. Reynolds number flow laminar □□□ turbulent □□□ □□□□□ identify □□□□□□□ □□□□□□□□□□□□□□.

$F = 6\pi\eta r v$ $Re = \rho v d/\eta$

Key points

- Honey has higher viscosity than water.
- Low Reynolds number indicates laminar flow.
- High Reynolds number indicates turbulent flow.

Application

Lubricants, pipelines, blood flow and hydraulic systems.

Study tip: Reynolds number has no unit.

Continuity equation and Bernoulli theorem

Lesson 5

ENGLISH

Continuity equation states that for steady incompressible flow, product of area and velocity remains constant. Bernoulli theorem states that pressure energy, kinetic energy and potential energy per unit volume have constant total along a streamline. Airfoil and atomiser work using Bernoulli principle.

MALAYALAM HELPER

Steady incompressible flow-□ area × velocity constant □□□ continuity equation. Bernoulli theorem □□□□□□□ streamline-□ pressure energy, kinetic energy, potential energy per unit volume total constant □□□. Airfoil, atomiser □□□□□□ Bernoulli principle □□□□□□□□□□□□□□.

$A_1v_1 = A_2v_2$ $P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$

Key points

- Narrow pipe gives higher velocity.
- Higher velocity region has lower pressure.
- Airfoil gets lift due to pressure difference.

Application

Atomiser, carburettor, airfoil, venturimeter and pipe flow.

Study tip: Fast fluid usually means low pressure.

Solved Problems

Exam-friendly method: Given Data → Formula Used → Substitution → Final Answer.

Problem 1

A force of 100 N acts on area 0.002 m^2. Find pressure.

$P = F/A = 100/0.002 = 50000$ Pa.

Problem 2

A wire extends 1 mm from original length 1 m. Find strain.

Strain $= \Delta L/L = 0.001/1 = 0.001$.

Problem 3

Pipe area changes from 4 cm^2 to 2 cm^2. If velocity in first part is 2 m/s, find second velocity.

$A_1v_1 = A_2v_2$. Therefore $v_2 = 4 \times 2 / 2 = 4$ m/s.

Problem 4

Find Re if $\rho = 1000$ kg/m^3, $v = 1$ m/s, $d = 0.02$ m, $\eta = 0.01$ Pa s.

$Re = \rho v d/\eta = 1000 \times 1 \times 0.02 / 0.01 = 2000$.

Expected Questions

Questions only are shown inside modules. Answers / hints are kept only in the Master Answer Key tab.

Q1. Define elasticity.

Q2. Define stress.

Q3. What is capillarity?

Q4. Define viscosity.

Q5. State Bernoulli theorem.

Long Answer Writing Practice

Use this structure for five-mark and ten-mark questions. Full answer checking is kept in the Master Answer Key tab.

State Bernoulli theorem and explain airfoil and atomiser.

- State theorem with formula.
- Mention conditions: steady, incompressible, non-viscous flow.
- Explain high velocity-low pressure relation.
- Apply to airfoil lift.
- Apply to atomiser spray.

Writing hint: Use the above points to build the answer in your own words.

1 mark
Definition or formula.

2 marks
Definition + one point.

5 marks
Statement + diagram + application.

10 marks
Detailed explanation + derivation + diagram.

Quick Revision Sheet

Last-minute exam checklist and common mistakes.

Final Master Answer Key

Answers are matched exactly with the Expected Questions shown inside each module.

Subject Code 1003

Applied Physics-I Complete Printable Notes

Author: Nandakumar.M • visit: polypmna.dpdns.org

Mini Formula Bank

Complete formula list for Applied Physics-I.

Physical quantity = numerical value × unit

Physical quantity

Every measurement must include a correct unit.

Absolute error = |measured value − true value|

Absolute error

Same unit as the measured quantity.

Relative error = absolute error / true value

Relative error

No unit; useful for comparing accuracy.

% error = relative error × 100

Percentage error

Write final value in percent.

$v = s / t$

Velocity

SI unit: m s^{-1}.

$a = (v - u) / t$

Acceleration

SI unit: m s^{-2}.

$v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$

Equations of motion

Use only for uniform acceleration.

$F = ma$

Force

Newton's second law; SI unit: newton.

$p = mv$

Momentum

SI unit: kg m s^{-1}.

$J = Ft = \Delta p$

Impulse

Impulse equals change in momentum.

$A_x = A \cos \theta$; $A_y = A \sin \theta$

Vector components

Resolve vector into horizontal and vertical components.

$R = \sqrt{(A^2 + B^2)}$

Resultant of perpendicular vectors

Use Pythagoras for 90° vectors.

$\omega = \theta / t$

Angular velocity

SI unit: rad s^{-1}.

$v = r\omega$; $a = r\alpha$

Linear-angular relation

Linear quantity = radius × angular quantity.

$a_c = v^2/r = r\omega^2$

Centripetal acceleration

Direction is towards the centre.

$F_c = mv^2/r$

Centripetal force

Required for circular motion.

$\tan\theta = v^2/(rg)$

Banking of roads

For no side friction condition.

$I = \Sigma mr^2;\ I = Mk^2$

Moment of inertia

Resistance to angular acceleration.

$I = I_G + Md^2$

Parallel axis theorem

Axis parallel to centre of gravity axis.

$I_z = I_x + I_y$

Perpendicular axis theorem

For plane lamina.

$\tau = rF\sin\theta$

Torque

Turning effect of force.

$L = I\omega$

Angular momentum

SI unit: kg m^2 s^{-1}.

$W = Fs\cos\theta$

Work

No displacement means no mechanical work.

$F = \mu N;\ N = mg$

Friction

On a horizontal surface.

$KE = \frac{1}{2}mv^2$

Kinetic energy

Energy due to motion.

$PE = mgh$

Potential energy

Energy due to position or height.

$P = W/t = E/t$

Power

SI unit: watt; 1 hp = 746 W.

$K = °C + 273;\ °F = (9/5)°C + 32$

Temperature conversion

Temperature rise in °C equals temperature rise in K.

$Q = mc\Delta T$

Heat

Specific heat relation.

$\sigma = F/A$

Stress

SI unit: pascal.

$strain = \Delta L/L$

Strain

No unit.

$Y = stress / strain$

Young's modulus

SI unit: pascal.

$P = F/A$

Pressure

Force per unit area.

$P_{abs} = P_{atm} + P_{gauge}$

Absolute pressure

Use absolute pressure in many fluid problems.

$h = 2T \cos\theta / (r\rho g)$

Capillary rise

Smaller tube radius gives greater rise.

$F = 6\pi\eta rv$

Stoke's law

Viscous force on a small sphere.

$Re = \rho vd/\eta$

Reynolds number

No unit; predicts nature of flow.

$A_1v_1 = A_2v_2$

Continuity equation

Incompressible steady flow.

$P + \frac{1}{2}\rho v^2 + \rho gh = constant$

Bernoulli theorem

Higher velocity gives lower pressure when height change is small.

Module 1 • 9 hours

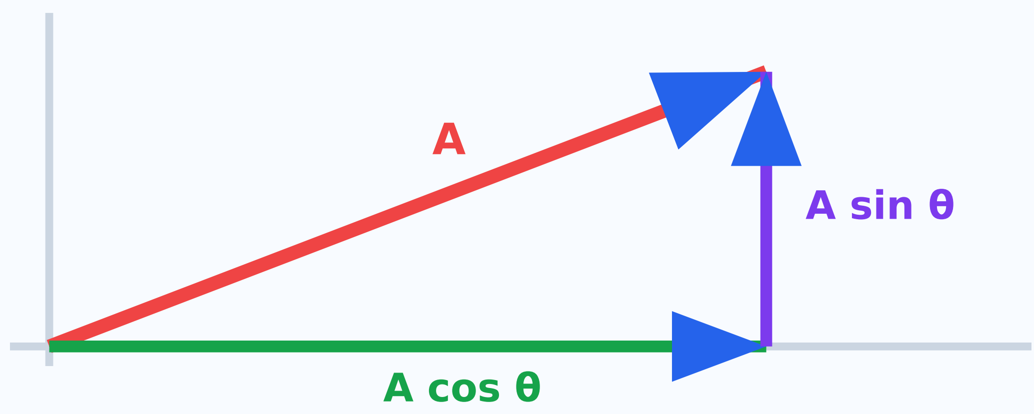
Measurements, Vectors and Laws of Motion

CO1: Apply laws of mechanics in rocket propulsion and recoil of gun.

□ module-□ units, errors, vectors, Newton's laws, momentum, impulse, recoil of gun, rocket propulsion □□□□□□ exam point of view-□ □□□□□□□□.

Visual diagram

Vector Resolution



Learning outcomes

- Physical quantities, units and systems
- Errors in measurement
- Scalar and vector quantities
- Equations of motion and Newton's laws
- Momentum, impulse, recoil and rocket propulsion

Physical quantities, units and systems

Lesson 1

ENGLISH

A physical quantity is anything that can be measured and expressed using a numerical value and a unit. Fundamental quantities are independent quantities such as length, mass and time. Derived quantities are obtained by combining fundamental quantities, for example velocity, acceleration, force and work.

MALAYALAM HELPER

[illegible] [illegible] [illegible] quantity [illegible] physical quantity. [illegible] numerical value [illegible] unit [illegible] [illegible]. Length, mass, time [illegible] fundamental quantities [illegible]. Velocity, acceleration, force [illegible] derived quantities [illegible].

Physical quantity = numerical value × unit | **$1\ N = 1\ kg\ m\ s^{-2}$** | **$1\ J = 1\ N\ m$**

Key points

- Use SI units in engineering answers.
- Write unit conversion clearly before substitution.
- Never write a numerical final answer without unit.

Application

Engineering drawings, laboratory readings and machine design calculations need consistent units.

Study tip: Number without unit is incomplete.

Errors in measurement

Lesson 2

ENGLISH

Error is the difference between measured value and true or accepted value. Systematic error is repeated in the same direction due to a faulty instrument or method. Random error changes unpredictably during repeated measurements.

MALAYALAM HELPER

Measured value-[illegible] true/accepted value-[illegible] [illegible] [illegible] error. Faulty instrument [illegible] wrong method [illegible] systematic error [illegible]. Repeated reading-[illegible] unpredictable variation [illegible] random error [illegible].

Absolute error = |measured − true| | **Relative error = absolute error / true value**

% error = relative error × 100

Key points

- Take repeated readings and mean value.
- Avoid parallax error while reading scale.
- Percentage error helps compare different measurements.

Application

Quality inspection, machining tolerance and laboratory experiments.

Study tip: Write absolute error with unit; relative error has no unit.

Scalar and vector quantities

Lesson 3

ENGLISH

A scalar quantity has magnitude only. A vector quantity has magnitude and direction. Temperature, mass and time are scalars. Displacement, velocity, acceleration, force and momentum are vectors. Vector resolution changes one vector into rectangular components.

MALAYALAM HELPER

Scalar quantity-□□□□ magnitude □□□□□□. Vector quantity-□□□□ magnitude □□□□ direction □□ □□□□. Temperature, mass, time scalar □□□. Displacement, velocity, acceleration, force, momentum vector □□□.

$A_x = A \cos \theta$ | $A_y = A \sin \theta$ | $R = \sqrt{(A^2 + B^2)}$

Key points

- Represent vector by an arrow.
- Arrow length shows magnitude.
- Arrow head shows direction.

Application

Crane lifting, bridge forces, slope motion and surveying.

Study tip: Perpendicular vectors form a right triangle.

Equations of motion and Newton's laws

Lesson 4

ENGLISH

Equations of motion connect displacement, velocity, acceleration and time for uniform acceleration. Newton's laws explain inertia, force and action-reaction. Force is the rate of change of momentum; for constant mass it becomes F = ma.

MALAYALAM HELPER

Uniform acceleration □□□□ motion-□ displacement, velocity, acceleration, time □□□□□□□□□□ □□□□□□□□ equations of motion. Newton's laws inertia, force, action-reaction explain □□□□□□□□□□. Constant mass □□□□□□□□ F = ma.

$v = u + at$ **$s = ut + \frac{1}{2}at^2$** **$F = ma$**

Key points

- First law explains inertia.
- Second law gives force equation.
- Third law states action and reaction are equal and opposite.

Application

Vehicle acceleration, braking distance and machine motion analysis.

Study tip: Use equations of motion only when acceleration is constant.

Momentum, impulse, recoil and rocket propulsion

Lesson 5

ENGLISH

Momentum is the product of mass and velocity. Impulse is the product of force and time and equals change in momentum. Recoil of gun and rocket propulsion are based on conservation of linear momentum. In a rocket, high-speed gases move backward and the rocket moves forward.

MALAYALAM HELPER

Momentum = mass × velocity. Impulse = force × time = change in momentum. Recoil of gun, rocket propulsion □□□□□□ conservation of linear momentum □□□□□□□□□□□□□□□□□□. Rocket-□ gases backward □□□□□□□□□ rocket forward move □□□□□□□.

$p = mv$ **$J = Ft = \Delta p$** **$m_1v_1 + m_2v_2 = \text{constant}$**

Key points

- Total momentum before and after interaction remains constant if external force is zero.
- Gun moves backward after bullet moves forward.
- Rocket does not need air to push against.

Application

Rockets, jet aircraft, pile driving and safety airbags.

Study tip: Impulse increases when stopping time increases; this reduces force.

Solved Problems

Exam-friendly method: Given Data → Formula Used → Substitution → Final Answer.

Problem 1

A force of 4 N acts on a 3 kg body. Find acceleration.

Given $F = 4$ N, $m = 3$ kg. Formula $a = F/m$. Therefore $a = 4/3 = 1.33$ m s^{-2}.

Problem 2

A 2 kg body moves at 5 m/s. Find momentum.

$p = mv = 2 \times 5 = 10$ kg m s^{-1}.

Problem 3

A force of 10 N acts for 0.5 s. Find impulse.

J = Ft = 10 × 0.5 = 5 N s.

Expected Questions

Questions only are shown inside modules. Answers / hints are kept only in the Master Answer Key tab.

Q1. Define physical quantity.

Q2. What is vector resolution?

Q3. State Newton's second law.

Q4. Why does a gun recoil?

Q5. Explain rocket propulsion.

Long Answer Writing Practice

Use this structure for five-mark and ten-mark questions. Full answer checking is kept in the Master Answer Key tab.

Explain conservation of linear momentum with recoil of gun.

- State conservation law.
- Before firing, total momentum is zero.
- After firing, bullet momentum forward equals gun momentum backward.
- This backward motion of gun is recoil.
- Add formula and application.

Writing hint: Use the above points to build the answer in your own words.

1 mark
Definition or formula.

2 marks
Definition + one point.

5 marks
Statement + diagram + application.

10 marks
Detailed explanation + derivation + diagram.

Module 2 • 9 hours

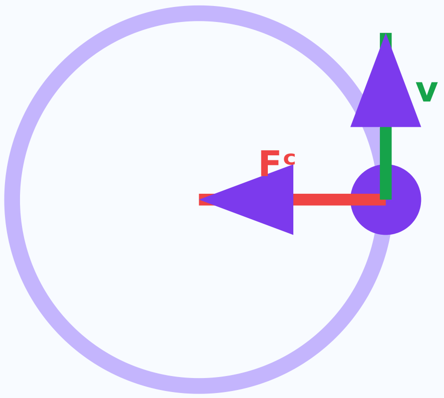
Circular and Rotational Motion

CO2: Apply concepts of circular and rotational motion.

Circular motion, angular quantities, centripetal force, banking of roads, moment of inertia, torque, angular momentum □□□□□□ □ module-□ □□□□□□□□.

Visual diagram

Circular Motion



Centripetal force acts towards the centre

Learning outcomes

- Angular displacement, velocity and acceleration
- Relation between linear and angular quantities
- Centripetal force and acceleration
- Banking of roads
- Moment of inertia, torque and angular momentum

Angular displacement, velocity and acceleration

Lesson 1

ENGLISH

Angular displacement is the angle swept by a rotating body. Angular velocity is rate of change of angular displacement. Angular acceleration is rate of change of angular velocity. Radian is used as the standard unit for angle in physics calculations.

MALAYALAM HELPER

Rotating body sweep □□□□□□□□□ angle □□□ angular displacement. Angular displacement-□□□□ rate □□□ angular velocity. Angular velocity-□□□□ rate □□□ angular acceleration. Calculation-□ radian □□□□□□□□□□□□□□□.

$\omega = \theta/t$ | $\alpha = (\omega - \omega_0)/t$ | $1 \text{ rev} = 2\pi \text{ rad}$

Key points

- One revolution = 2π rad.
- Angular velocity is measured in rad s^{-1}.
- Angular acceleration is measured in rad s^{-2}.

Application

Fans, wheels, turbines and rotating shafts.

Study tip: Convert rpm to rad/s before numerical problems.

Relation between linear and angular quantities

Lesson 2

ENGLISH

A point on a rotating body has both angular motion and linear motion. Linear velocity depends on radius and angular velocity. Points farther from the centre have greater linear velocity for the same angular velocity.

MALAYALAM HELPER

Rotating body-□ □□□□ point-□□□□ angular motion □□ linear motion □□ □□□□□□□□. Linear velocity radius-□□□ angular velocity-□□□□□ proportional □□□. Centre-□ □□□□□□ □□□□ □□□□□□□□□ linear velocity □□□□□.

$s = r\theta$ | $v = r\omega$ | $a = r\alpha$

Key points

- All points in a rigid body have same angular velocity.
- Outer points travel more distance in same time.

- Tangential acceleration is related to angular acceleration.

Application

Wheel rims, pulley belts and gear systems.

Study tip: The letter r connects circular and linear quantities.

Centripetal force and acceleration

Lesson 3

ENGLISH

A body moving in a circle needs acceleration towards the centre even if speed is constant. This acceleration is centripetal acceleration. The force producing it is centripetal force. Without this force, the body moves tangentially.

MALAYALAM HELPER

Constant speed-□ circle-□ move □□□□□□□□□ direction □□□□□□□□□□□□ acceleration centre direction-□ □□□□□□□□. □□□□□ centripetal acceleration. □□□□□ □□□□□□□□ force centripetal force □□□.

$a_c = v^2/r$ | $a_c = r\omega^2$ | $F_c = mv^2/r$

Key points

- Direction is always towards centre.
- It changes direction of velocity.
- It is not a new force; it is provided by tension, friction, gravity etc.

Application

Vehicle turning, satellites, stone tied to a string.

Study tip: Centripetal means centre-seeking.

Banking of roads

Lesson 4

ENGLISH

Banking means raising the outer edge of a curved road. It provides the required centripetal force safely while vehicles take a turn. Proper banking reduces dependence on friction and improves safety at curves.

MALAYALAM HELPER

Curved road-□□□□ outer edge □□□□□□□□□□□□□□□ banking. Vehicle turn □□□□□□□□□□□ □□□□□ centripetal force safely provide □□□□□□□ □□□ □□□□□□□□□□□□□□. Banking friction dependence □□□□□□□□□□□□□□.

$\tan \theta = v^2/(rg)$ $v = \sqrt{rg \tan \theta}$

Key points

- Outer edge is higher than inner edge.
- Useful on highways and railway tracks.
- Design depends on speed and radius of curve.

Application

Road curves, railway tracks and race tracks.

Study tip: Higher design speed needs greater banking angle.

Moment of inertia, torque and angular momentum

Lesson 5

ENGLISH

Moment of inertia is the rotational analogue of mass. It depends on mass and distribution of mass about the axis of rotation. Torque is turning effect of force. Angular momentum is the product of moment of inertia and angular velocity.

MALAYALAM HELPER

Moment of inertia rotational motion-□□ mass □□□□□□□□. Mass □□□□□□ axis-□ □□□□□□□ distribute □□□□□□□□□□□□□□□□ □□□□□□□□ □□□□□□□□□□□□. Torque force-□□□□ turning effect □□□. Angular momentum = $I\omega$.

$I = \Sigma m r^2$ | $\tau = I\alpha$ | $L = I\omega$

Key points

- More mass away from axis gives more moment of inertia.
- Torque produces angular acceleration.
- Angular momentum is conserved when external torque is zero.

Application

Flywheels, turbines, gyroscopes and rotating machines.

Study tip: Moment of inertia increases strongly with radius because of r^2.

Solved Problems

Exam-friendly method: Given Data → Formula Used → Substitution → Final Answer.

Problem 1

A wheel has radius 0.5 m and angular velocity 4 rad/s. Find linear velocity.

$v = r\omega = 0.5 \times 4 = 2$ m/s.

Problem 2

A 2 kg body moves in a circle of radius 1 m with speed 4 m/s. Find centripetal force.

$F = mv^2/r = 2 \times 16 / 1 = 32$ N.

Problem 3

A body has $I = 3$ kg m² and $\omega = 5$ rad/s. Find angular momentum.

$L = I\omega = 3 \times 5 = 15$ kg m^2 s^{-1}.

Expected Questions

Questions only are shown inside modules. Answers / hints are kept only in the Master Answer Key tab.

Q1. Define angular velocity.

Q2. What is centripetal force?

Q3. Why are roads banked?

Q4. Define moment of inertia.

Q5. State parallel axis theorem.

Long Answer Writing Practice

Use this structure for five-mark and ten-mark questions. Full answer checking is kept in the Master Answer Key tab.

Explain banking of roads.

- Define banking.
- Draw a vehicle on a banked curve.
- Mention centripetal force requirement.
- Write formula $\tan \theta = v^2/(rg)$.
- Give road and railway applications.

Writing hint: Use the above points to build the answer in your own words.

1 mark
Definition or formula.

2 marks
Definition + one point.

5 marks
Statement + diagram + application.

10 marks
Detailed explanation + derivation + diagram.

Module 3 • 9 hours

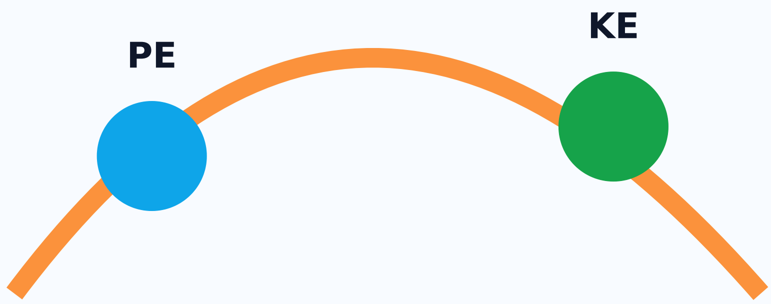
Work, Energy, Power, Friction and Heat

CO3: Solve problems involving work, energy, power, temperature and friction.

Work, friction, energy, power, conservation of energy, heat, temperature, thermometry, heat transfer □□□□□□ □ module-□ □□□□□□□□.

Visual diagram

Energy Transformation



Potential energy changes into kinetic energy

Learning outcomes

- Work and friction
- Energy and conservation of energy
- Power and efficiency
- Heat, temperature and thermometry
- Modes of heat transfer

Work and friction

Lesson 1

ENGLISH

Work is done when a force produces displacement. Work depends on force, displacement and angle between them. Friction is a resistive force that opposes relative motion. Static friction acts before motion starts and kinetic friction acts during motion.

MALAYALAM HELPER

Force displacement produce □□□□□□□ work done □□□□□ □□□□□□□□. Work force, displacement, angle □□□□□□□□ □□□□□□□□□□□. Relative motion-□□ oppose □□□□□□□□□ resistive force □□□ friction. Motion start □□□□□□□□□□□□□ □□□□□□ static friction, motion □□□□□□□□□□□ kinetic friction.

W = Fs cos θ | **F = μN** | **N = mg**

Key points

- No displacement means no mechanical work.
- Friction helps walking and braking.
- Friction wastes energy as heat in machines.

Application

Brakes, clutches, belt drives and lubrication.

Study tip: Friction is useful and harmful depending on situation.

Energy and conservation of energy

Lesson 2

ENGLISH

Energy is the capacity to do work. Kinetic energy is energy due to motion. Potential energy is energy due to position. According to conservation of energy, energy can neither be created nor destroyed; it can only be transformed from one form to another.

MALAYALAM HELPER

Work □□□□□□□□□□□ capacity □□□ energy. Motion □□□□□ □□□□ energy kinetic energy. Position/height □□□□□ □□□□ energy potential energy. Conservation of energy □□□□□□□ energy create □□□□□□□□ destroy □□□□□□□□ □□□□□□□□; □□□ form-□ □□□□□□ □□□□□□□ form-□□□□□□ □□□□□.

KE = ½mv² | **PE = mgh** | **Total energy = constant**

Key points

- Falling body converts PE into KE.
- Machines convert energy into useful work.
- Losses usually appear as heat and sound.

Application

Hydroelectric power, cranes, lifts and machine tools.

Study tip: In falling body problems, PE decreases and KE increases.

Power and efficiency

Lesson 3

ENGLISH

Power is the rate of doing work or rate of energy conversion. A machine with higher power performs the same work in less time. Efficiency compares useful output energy or power with input energy or power.

MALAYALAM HELPER

Work □□□□□□□□□□□□□□□□ rate □□□□□□□□□□□ energy conversion-□□□□ rate □□□ power. Higher power machine same work □□□□□□□□ time-□ □□□□□□□□. Efficiency useful output-□□ input-□□□□□ compare □□□□□□□□□□□.

$P = W/t$ **$P = E/t$** **Efficiency = output/input × 100**

Key points

- Power unit is watt.
- Commercial unit of electrical energy is kWh.
- Efficiency is always less than 100% for practical machines.

Application

Motors, pumps, lifts and power plants.

Study tip: 1 horsepower = 746 W.

Heat, temperature and thermometry

Lesson 4

ENGLISH

Heat is energy transferred because of temperature difference. Temperature indicates degree of hotness or coldness. Thermometers measure temperature using a physical property that changes uniformly with temperature. Mercury thermometer and pyrometer are common examples.

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Temperature difference □□□□□ transfer □□□□□□□□□ energy □□□ heat. Temperature hot/cold level □□□□□□□□□□□□. Temperature change □□□□□□□□□□ physical property uniformly □□□□□□□□□ □□□□□□□□□□ thermometer measure □□□□□□□□□□.

$Q = mc\Delta T$ | $K = °C + 273$ | $°F = 1.8°C + 32$

Key points

- Heat unit is joule.
- Temperature unit is kelvin in SI.
- Pyrometer measures high temperature without contact.

Application

Furnaces, boilers, engine cooling and heat exchangers.

Study tip: Do not write °K; correct SI symbol is K.

Modes of heat transfer

Lesson 5

ENGLISH

Heat is transferred by conduction, convection and radiation. Conduction needs material contact. Convection occurs through bulk movement of fluid. Radiation transfers heat by electromagnetic waves and can travel through vacuum.

MALAYALAM HELPER

Heat transfer conduction, convection, radiation □□□□□ modes □□□ □□□□□□□□□□□. Conduction-□□ contact □□□□. Convection fluid movement □□□. Radiation vacuum-□ □□□□ travel □□□□□□□.

$Q = mc\Delta T$ **Heat flows from high temperature to low temperature**

Key points

- Metal spoon gets hot by conduction.
- Sea breeze is due to convection.
- Sunlight reaches Earth by radiation.

Application

Insulation, furnaces, radiators and solar heating.

Study tip: Radiation does not need a medium.

Solved Problems

Exam-friendly method: Given Data → Formula Used → Substitution → Final Answer.

Problem 1

A force of 20 N moves a body 6 m in same direction. Find work.

$W = Fs = 20 \times 6 = 120$ J.

Problem 2

Find KE of 2 kg body moving at 5 m/s.

$KE = \frac{1}{2}mv^2 = 0.5 \times 2 \times 25 = 25$ J.

Problem 3

Calculate power if 400 J work is done in 4 s.

P = W/t = 400/4 = 100 W.

Problem 4

Convert 100°C to Fahrenheit.

F = 1.8C + 32 = 180 + 32 = 212°F.

Expected Questions

Questions only are shown inside modules. Answers / hints are kept only in the Master Answer Key tab.

Q1. Define work.

Q2. What is friction?

Q3. State conservation of energy.

Q4. Define power.

Q5. Name modes of heat transfer.

Long Answer Writing Practice

Use this structure for five-mark and ten-mark questions. Full answer checking is kept in the Master Answer Key tab.

Explain modes of heat transfer with examples.

- Define heat transfer.
- Explain conduction with example.
- Explain convection with example.
- Explain radiation with example.
- Add engineering applications.

Writing hint: Use the above points to build the answer in your own words.

1 mark
Definition or formula.

2 marks
Definition + one point.

5 marks
Statement + diagram + application.

10 marks
Detailed explanation + derivation + diagram.

Module 4 • 9 hours

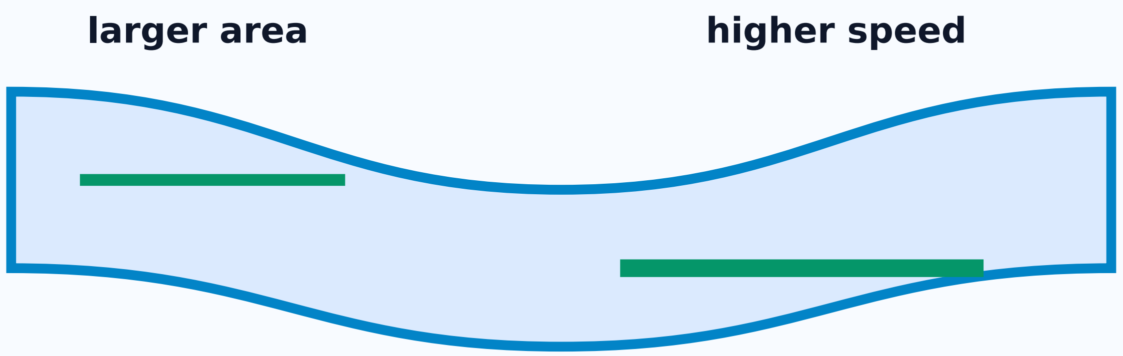
Elasticity, Fluids and Bernoulli Applications

CO4: Use fluid dynamics in atomiser and airfoil.

Elasticity, stress, strain, pressure, surface tension, viscosity, continuity equation, Bernoulli theorem, airfoil, atomiser □□□□□□ □□□□□□□□.

Visual diagram

Continuity and Bernoulli



Narrow region: velocity increases, pressure decreases

Learning outcomes

- Elasticity, stress and strain
- Pressure and fluids
- Surface tension and capillarity
- Viscosity and Reynolds number
- Continuity equation and Bernoulli theorem

Elasticity, stress and strain

Lesson 1

ENGLISH

Elasticity is the property by which a material regains its original shape and size after removal of deforming force. Stress is restoring force per unit area. Strain is the ratio of change in dimension to original dimension. Young's modulus is stress divided by strain.

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Deforming force remove □□□□□□□ original shape and size regain □□□□□□□□□□□ property □□□ elasticity. Stress = restoring force per unit area. Strain = change in dimension / original dimension. Young's modulus = stress / strain.

Stress = F/A **Strain = ΔL/L** **Y = stress/strain**

Key points

- Stress has unit pascal.
- Strain has no unit.
- Hooke's law is valid within elastic limit.

Application

Springs, bridges, beams, wires and machine components.

Study tip: Stress has unit; strain has no unit.

Pressure and fluids

Lesson 2

ENGLISH

Pressure is force per unit area. In fluids, pressure acts normally on any surface. Gauge pressure is pressure above atmospheric pressure. Absolute pressure is the sum of atmospheric pressure and gauge pressure.

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Pressure = force per unit area. Fluid pressure surface-□ normal direction-□ act □□□□□□□□. Gauge pressure atmospheric pressure-□□□□□□□□ □□□□□ □□□□ pressure □□□. Absolute pressure = atmospheric pressure + gauge pressure.

$P = F/A$ **$P = \rho gh$** **$P_{abs} = P_{atm} + P_{gauge}$**

Key points

- Pressure unit is pascal.
- Smaller area gives greater pressure for same force.

- Fluid pressure increases with depth.

Application

Hydraulic machines, dams, pressure gauges and pumps.

Study tip: Use absolute pressure when total pressure is required.

Surface tension and capillarity

Lesson 3

ENGLISH

Surface tension is the property of a liquid surface due to which it behaves like a stretched elastic membrane. Capillarity is the rise or fall of liquid in a narrow tube due to surface tension and adhesive/cohesive forces.

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Liquid surface stretched membrane □□□□ behave □□□□□□□□□□□ property □□□ surface tension. Narrow tube-□ liquid rise/fall □□□□□□□□□□□ capillarity □□□. Surface tension, adhesive/cohesive forces □□□□□ □□□ □□□□□□□□□□□□□□.

$h = 2T \cos\theta / (r\rho g)$ | **Surface tension unit = N/m**

Key points

- Small drops are nearly spherical due to surface tension.
- Capillary rise increases when tube radius decreases.
- Meniscus shape depends on liquid and tube material.

Application

Ink pens, wicks, soil water movement and medical capillary tubes.

Study tip: Narrower capillary tube gives greater rise.

Viscosity and Reynolds number

Lesson 4

ENGLISH

Viscosity is the internal resistance offered by a fluid to relative motion between its layers. A highly viscous liquid flows slowly. Reynolds number is a dimensionless number used to identify laminar or turbulent flow.

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Fluid layers □□□□□□□□□□ relative motion-□□ oppose □□□□□□□□□ internal resistance □□□ viscosity. Highly viscous liquid slow □□□ flow □□□□□□□. Reynolds number flow laminar □□□ turbulent □□□ □□□□□ identify □□□□□□□ □□□□□□□□□□□□□□.

$F = 6\pi\eta rv$ **$Re = \rho vd/\eta$**

Key points

- Honey has higher viscosity than water.
- Low Reynolds number indicates laminar flow.
- High Reynolds number indicates turbulent flow.

Application

Lubricants, pipelines, blood flow and hydraulic systems.

Study tip: Reynolds number has no unit.

Continuity equation and Bernoulli theorem

Lesson 5

ENGLISH

Continuity equation states that for steady incompressible flow, product of area and velocity remains constant. Bernoulli theorem states that pressure energy, kinetic energy and potential energy per unit volume have constant total along a streamline. Airfoil and atomiser work using Bernoulli principle.

MALAYALAM HELPER

Steady incompressible flow-□ area × velocity constant □□□ continuity equation. Bernoulli theorem □□□□□□□ streamline-□ pressure energy, kinetic energy, potential energy per unit volume total constant □□□. Airfoil, atomiser □□□□□□ Bernoulli principle □□□□□□□□□□□□□□.

$A_1v_1 = A_2v_2$ **$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$**

Key points

- Narrow pipe gives higher velocity.
- Higher velocity region has lower pressure.
- Airfoil gets lift due to pressure difference.

Application

Atomiser, carburettor, airfoil, venturimeter and pipe flow.

Study tip: Fast fluid usually means low pressure.

Solved Problems

Exam-friendly method: Given Data → Formula Used → Substitution → Final Answer.

Problem 1

A force of 100 N acts on area 0.002 m². Find pressure.

$P = F/A = 100/0.002 = 50000$ Pa.

Problem 2

A wire extends 1 mm from original length 1 m. Find strain.

Strain = $\Delta L/L = 0.001/1 = 0.001$.

Problem 3

Pipe area changes from 4 cm² to 2 cm². If velocity in first part is 2 m/s, find second velocity.

$A_1v_1 = A_2v_2$. Therefore $v_2 = 4 \times 2 / 2 = 4$ m/s.

Problem 4

Find Re if $\rho = 1000$ kg/m³, $v = 1$ m/s, $d = 0.02$ m, $\eta = 0.01$ Pa s.

$Re = \rho v d/\eta = 1000 \times 1 \times 0.02 / 0.01 = 2000$.

Expected Questions

Questions only are shown inside modules. Answers / hints are kept only in the Master Answer Key tab.

Q1. Define elasticity.

Q2. Define stress.

Q3. What is capillarity?

Q4. Define viscosity.

Q5. State Bernoulli theorem.

Long Answer Writing Practice

Use this structure for five-mark and ten-mark questions. Full answer checking is kept in the Master Answer Key tab.

State Bernoulli theorem and explain airfoil and atomiser.

- State theorem with formula.
- Mention conditions: steady, incompressible, non-viscous flow.
- Explain high velocity-low pressure relation.
- Apply to airfoil lift.
- Apply to atomiser spray.

Writing hint: Use the above points to build the answer in your own words.

1 mark
Definition or formula.

2 marks
Definition + one point.

5 marks
Statement + diagram + application.

10 marks
Detailed explanation + derivation + diagram.

Revision

Last-minute checklist.

Common mistakes

- Missing SI units.
- Wrong unit conversion.
- Confusing scalar and vector.
- Using wrong formula.

Important diagrams

- Vector resolution.
- Banking of roads.
- Energy transformation.
- Bernoulli pipe and airfoil.

Answer pattern

- Definition.
- Formula with unit.
- Diagram.
- Application.

Master Answer Key

Answers are matched exactly with the Expected Questions in each module.

Module 1: Measurements, Vectors and Laws of Motion

No.	Question	Answer / Hint
Q1	Define physical quantity.	**A measurable quantity expressed with numerical value and unit.**
Q2	What is vector resolution?	**Splitting a vector into perpendicular components.**
Q3	State Newton's second law.	**Rate of change of momentum is directly proportional to applied force and occurs in the direction of force.**
Q4	Why does a gun recoil?	**Due to conservation of linear momentum.**
Q5	Explain rocket propulsion.	**Rocket gases move backward at high speed; equal and opposite reaction pushes rocket forward.**

Long answer model

Question: Explain conservation of linear momentum with recoil of gun.

Model answer: When no external force acts on a system, total linear momentum remains constant. Before firing, gun and bullet are at rest, so total momentum is zero. After firing, the bullet moves forward with momentum and the gun moves backward with equal opposite momentum. Therefore total momentum remains zero. This backward motion of the gun is called recoil.

Module 2: Circular and Rotational Motion

No.	Question	Answer / Hint
Q1	Define angular velocity.	**Rate of change of angular displacement.**
Q2	What is centripetal force?	**Force acting towards the centre that keeps a body in circular motion.**
Q3	Why are roads banked?	**To provide required centripetal force safely at curves.**
Q4	Define moment of inertia.	**Rotational inertia of a body about an axis.**
Q5	State parallel axis theorem.	**$I = I^G + Md^2$.**

Long answer model

Question: Explain banking of roads.

Model answer: Banking of road is the raising of outer edge of a curved road above the inner edge. When a vehicle moves through a curve, centripetal force is required towards the centre.

In a properly banked road, the horizontal component of normal reaction helps to provide this centripetal force. Hence the vehicle can take the turn safely with less dependence on friction.

Module 3: Work, Energy, Power, Friction and Heat

No.	Question	Answer / Hint
Q1	Define work.	**Work is done when force produces displacement.**
Q2	What is friction?	**Resistive force opposing relative motion.**
Q3	State conservation of energy.	**Energy can neither be created nor destroyed, only transformed.**
Q4	Define power.	**Rate of doing work.**
Q5	Name modes of heat transfer.	**Conduction, convection and radiation.**

Long answer model

Question: Explain modes of heat transfer with examples.

Model answer: Heat flows from a body at higher temperature to a body at lower temperature. In conduction, heat is transferred through direct contact, as in a metal spoon kept in hot tea. In convection, heat is transferred by movement of fluid, as in sea breeze. In radiation, heat is transferred by electromagnetic waves and no medium is needed; sunlight reaching Earth is an example.

Module 4: Elasticity, Fluids and Bernoulli Applications

No.	Question	Answer / Hint
Q1	Define elasticity.	**Ability to regain original shape and size after removal of deforming force.**
Q2	Define stress.	**Restoring force per unit area.**
Q3	What is capillarity?	**Rise or fall of liquid in a narrow tube due to surface tension.**
Q4	Define viscosity.	**Internal resistance of a fluid to relative motion between layers.**
Q5	State Bernoulli theorem.	**For steady incompressible non-viscous flow, $P + \frac{1}{2}\rho v^2 + \rho g h$ remains constant along a streamline.**

Long answer model

Question: State Bernoulli theorem and explain airfoil and atomiser.

Model answer: Bernoulli theorem states that for steady, incompressible and non-viscous flow along a streamline, $P + \frac{1}{2}\rho v^2 + \rho g h$ remains constant. When height change is small, higher velocity causes lower pressure. In an airfoil, air moves faster above the curved upper surface, so pressure above becomes lower and lift is produced. In an atomiser, fast air over a tube lowers pressure, so liquid rises and sprays into fine droplets.